

Tuesday Warm Up: Chapters 4-5

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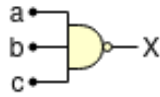


Figure 1: A 3-input NAND gate is LOW only when all three inputs are HIGH.

expression, create the Karnaugh map and the truth table. (c) Generate the simplified Boolean expression from the Karnaugh map. *Hint: remember that cells in the top row are adjacent to those in the bottom row.*

1 Logic Circuits, Boolean Algebra, and Karnaugh Maps

1. Consider the three input NAND gate in Fig. 1. (a) Based on Fig. 1, create the truth table for the 3 input NAND. How many terms would there be in the S-SOP expression for this circuit? (b) From the data in the truth table, create the 3-variable Karnaugh map. (c) Generate the simplified Boolean expression from the Karnaugh map. *Hint: remember that cells in the top row are adjacent to those in the bottom row.*
4. Draw the simplified logic function using basic gates.

2. Draw the simplified logic function using basic gates.

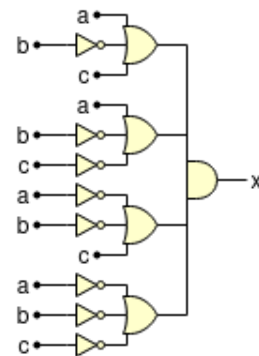


Figure 2: A complex logic function.

3. Consider the logic function in Fig. 2. (a) Write an S-POS expression for the output. (b) From the S-POS